

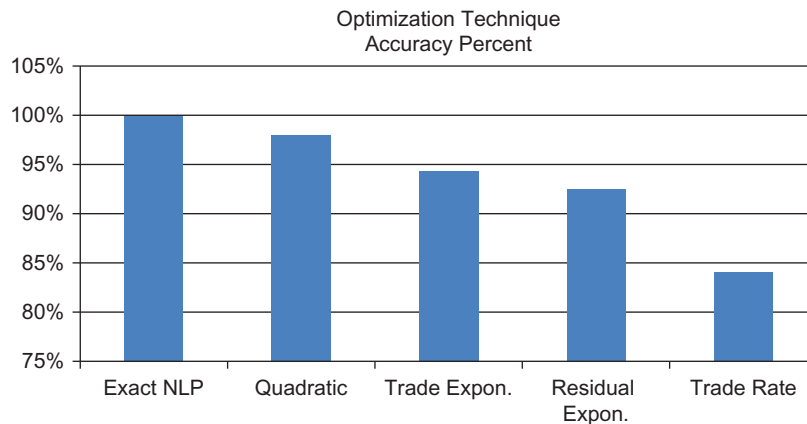
(QP) provided dramatic improvement over the non-linear approach and only took 4.4 minutes to solve. But for a trader, even 4.4 minutes may be too long especially if they desire to perform re-optimization during the day. The fastest solutions for a 500-stock portfolio were the 14 seconds for the trade schedule exponential solution, 10 seconds for the residual schedule exponential, and only 5 seconds for the trade rate technique. These are all dramatic time improvements over the NLP and QP formulations.

*How accurate was the solution for each optimization technique?*

Figure 9.2 shows the accuracy of each approach. The quadratic optimizer was 98% accurate. This was followed by the trade schedule exponential 93%, residual schedule exponential 91%, and trade rate technique at 84%. Accuracy was measured as 1 minus the error between the actual trade schedule determined from the NLP solution and the trade schedule determined from each of our approaches.

This simulation experiment highlighted the inverse relationship between solution time and accuracy. The quicker we solve, the less accurate the solution. In some circles this has become known as the “developer’s dilemma.” Solving too fast may give an inaccurate result, but solving too slow may miss the opportunity altogether.

It appears that the exponential approaches provide the highest level of accuracy and the quickest solution times. Additionally, the exceptionally quick solution of the trade rate technique could be used in conjunction with the exponential approaches or quadratic optimization for full risk management



■ Figure 9.2 Optimization Technique Accuracy Percent.